STI COLLEGE STA. MESA • BS INFORMATION TECHNOLOGY

Information & Communications Technology Program • Academic Year 2026–2027

ECO QUEST

A Game That Practices Proper Segregation and Teaches the Consequences of Littering

OFFICIAL DEFENSE PREPARATION GUIDE & 100 PANEL QUESTIONS

100	44	35	21
TOTAL QUESTIONS	HARD / DEFENSE TRAPS	MEDIUM DEPTH	FOUNDATIONAL EASY

PROPONENTS & TECHNICAL ROLES

Bryan Duarte: Lead Architecture, Physics Raycasting, C# Quests

Marianne Rouze B. Pillado: 2D UI/UX Design, Dialogue Copy, Quiz Data

Marjorie R. Pamonag: 3D Scene Assembly, Environment Lighting, URP

Joanna Niña C. Tagaca: Requirements Matrix, ISO 25010, Beta Testing

Adviser: Mark Angelo C. De Guzman | **Coordinator:** Chris Jericho M. Ocampo | **Lead Panelist:** Miguel Jerome Silverio

Target Defense Date: **November 14, 2026** | Strict Read-Only Verification Build: **Unity 6000.3.11f1 (URP)**

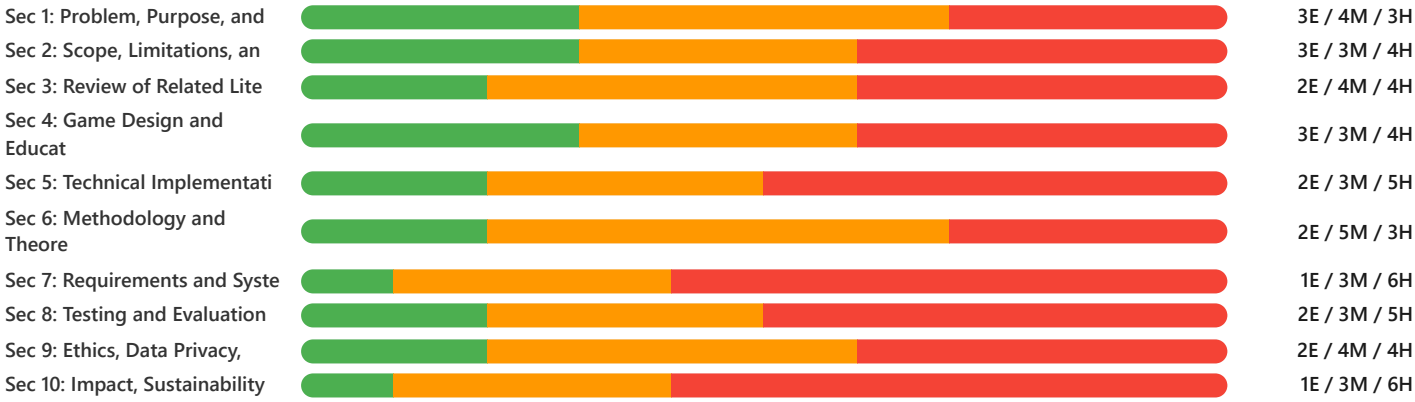
How to Use This Preparation Guide — Methodology & Overview

Defense Coaching Instructions: Practice answering these questions *out loud in your own words*. Do NOT memorize text verbatim. The panel values technical authenticity, understanding of real code mechanics, and honest limitation awareness far above recited speeches. Allocate 60% of team study time to the **HARD** questions.

Legend & Question Difficulty Breakdown

Tag	Meaning	Candidate Strategy & Delivery Guidance
EASY	Foundational Concepts	Answer decisively in 1 to 2 clear sentences. State facts cleanly without over-explaining.
MEDIUM	System Connections	Explain how two parts interact (e.g. DFD process to data store, or quiz data to currency rewards).
HARD	Defense Traps & Weaknesses	**Never argue against a true limitation.** Name the weakness first, explain why the trade-off was made, and explain how Version 2.0 solves it.
CONFIRM	Runtime Playtest Check	Items marked CONFIRM require the team to verify the runtime behavior in the Unity Game view before defending.

Question Distribution by Defense Section (10 Questions Each)



Rule of the Defense Room: When a panelist identifies a missing feature or documentation error, state: "*Yes, panelist, that is an acknowledged boundary in the current build. We documented this trade-off on page X, and our technical upgrade path for Version 2.0 is Y.*" This disarms hostility and demonstrates academic maturity.

Game Technical Verification Summary — Step 0 Codebase Facts

The following technical facts have been directly verified by inspecting the Unity project files on disk (scripts, scenes, prefabs, and ScriptableObjects):

Subsystem / Metric	Code Evidence Path	Verified Runtime Behavior & Specification
Engine & Pipeline	`ProjectSettings/ProjectVersion.txt`	Unity **6000.3.11f1** (URP 17.6.0). Target: Windows Standalone (x86_64). Input System 1.20.0.
Active Scenes in Build	`EditorBuildSettings.asset`	Index 0: `Main Menu.unity`, Index 1: `Forest Scene.unity`, Index 2: `City Scene.unity`. (Beach.unity is unlinked on disk).
Waste Categories	`QuestManager.cs`, `CityTrashDatabase.asset`	**Forest:** 3 physical bins (Biodegradable, Recyclable, NonBiodegradable). **City:** 4 categories (+Special/Hazard).
Verified Forest Items	`Assets/EcoQuest/Forest Scene.unity`	9 physical trash prefabs: banana peel, fish bone, food scraps (Bio); cans, paper (Recyc); plastic cup (Non-Bio).
Drag & Drop Physics	`DragAndDrop.cs`	Raycast + SphereCast (0.12m radius, 5m range). Attached via kinematic socket lerp (speed 25). Throw force 3.5 impulse.
Penalty Mechanics	`QuestManager.HandleMistake()`	1st & 2nd mistake = warning banner + ejection. 3rd mistake = **-10 Eco-Coins** deducted (clamped at 0). Ground drop = **0 penalty**.
Educational Quizzes	`QuizManager.cs`, `QuizData.asset`	6 identical questions shuffled via Fisher-Yates. Pre-Quiz has zero feedback; Post-Quiz awards +5 gold/correct & shows learning gain.
Eco Shop Catalog	`UI_EcoShop.cs`, `InventoryItemDatabase`	9 consumable items (15 to 55 coins) restoring Hunger (+10% to +50%) and Stamina buffs. **Zero waste spawned on consumption.**
Daily Quests & Reset	`CityTrashDetectiveManager.cs`	**Cooldown timers**, NOT midnight resets: Ranger Robin (5 mins), Guardian Quiz (10 mins). System clock change bypasses cooldown.
LitterBug Pedestrian	`LitterbugQuestManager.cs`	Civilian pedestrian walks sidewalks, throws litter props within 20m. Player persuades within 2.4m via [E]; NPC apologizes and stops.
Save & Storage System	`PlayerPrefs` (Windows Registry)	Saved in `HKCU\Software\DefaultCompany\EcoQuest` as unencrypted plaintext key-values. No hashing. Missing keys return 0 safely.
Unscoped Live Systems	`CityTrafficManager.cs`, `PlayerNeeds.cs`	24-slot inventory [B], Hunger decay & Stamina vitals, autonomous city traffic, crowd pedestrian walk graphs.

Priority Manuscript Fixes Before Defense — Critical Panelist Traps

The following 8 discrepancies must be acknowledged by the team. Use the suggested verbal defenses during the presentation:

Priority	Discrepancy / Trap	What to Say in the Defense (Action & Scripted Answer)	Rel. Q#
HIGH	**12 In-Text Citations Missing from References**	State: "*During final document assembly, 12 cited literature sources were omitted from the bibliography. We have compiled the complete APA 7th edition addendum ready for immediate insertion.*"	Q23
HIGH	**Raw File Names in References List**	Acknowledge that PDF file names were pasted as placeholders. State the full formal citations for ISWA (2021) and Satria et al. (2021).	Q25
HIGH	**Beta Test Q1: 42.6% answered Plastic Bottle as Bio**	State: "*This finding reveals a deep real-world pupil misconception—confusing recyclable with biodegradable. It directly proves why Eco Quest is necessary, and shows where our tutorial hints must be reinforced.*"	Q72
HIGH	**Waste Categories: Paper says 2, Code has 3 & 4**	State: "*The proposal draft scoped Bio and Non-Bio, but implementation adopted the standard Philippine 3-bin system (adding Recyclable) and City e-waste for curriculum completeness.*"	Q12
HIGH	**Objective 3 promises Behavior Change**	Admit: "*A cross-sectional 15-minute game measures knowledge acquisition and situational awareness. Real-world behavioral change requires longitudinal tracking, which is noted in our Limitations.*"	Q3
HIGH	**ISO 25010 Confidentiality Overclaim**	Admit: "*PlayerPrefs stores plaintext values in Windows Registry. While school non-admin accounts prevent pupil access, Version 2.0 will implement AES-256 save encryption.*"	Q65
MED	**Daily Quest is Cooldown-Based**	Explain: "*To facilitate retesting in single class periods, quests use 5- and 10-minute cooldowns rather than 24-hour midnight locks, enabling pupils to replay scenarios during laboratory time.*"	Q17
MED	**Blank Tables in Manuscript**	Acknowledge template export omission for Implementation Plan and Specific Requirements; provide the completed 5-phase schedule addendum to the panel.	Q80

Verify by Playing Checklist — Pre-Defense Playtest Verification

Code reading verifies syntax, but only playing verifies user experience. The team must verify these 8 items in the Unity Game view before stepping into the defense room:

Check	Item to Verify	Exact Playtest Verification Action	Expected Result
[ ]	**Guardian Flow & Pre-Quiz**	Start new game in Forest Scene. Move through tutorial. Approach Guardian NPC and press [E].	Dialogue opens, Cinemachine blends camera, Pre-Quiz modal displays 6 questions.
[ ]	**Wrong Trash Banner**	Pick up a banana peel, walk to the Blue Recyclable Bin, and release it.	Item is ejected out with impulse; WrongTrashPopup banner slides down: "💡 Hint: Put this in the Green Bin! • Mistake 1/3".
[ ]	**3-Mistake Penalty**	Intentionally throw 3 wrong items consecutively into incorrect bins.	On 3rd mistake, 10 Eco-Coins are deducted from the coin counter and a warning popup displays.
[ ]	**Post-Quiz & Learning Gain**	Sort all 9 trash items correctly into the 3 bins.	Post-Quiz opens, shows feedback, awards +5 coins per correct answer, displays Learning Gain scorecard.
[ ]	**Gate Transition to City**	Approach Explorer NPC at stone archway and press [E].	Dialogue confirms Forest complete; scene transition loads City Scene asynchronously.
[ ]	**Ranger Robin Checklist [F]**	In City Scene, talk to Ranger Robin in the park. Accept the quest and press [F].	HUD timer starts (4:00), compass arrow points toward nearest trash with distance in meters.
[ ]	**LitterBug Persuasion**	Locate pedestrian throwing litter with overhead '!' indicator. Approach within 2.4m and press [E].	Persuasion dialogue opens; NPC apologizes, becomes Persuaded, and displays green checkmark.
[ ]	**Eco Shop Hunger Restore**	Sprint until Stamina/Hunger depletes. Talk to Handyman, buy an Apple for 15 coins, open Bag [B], use item.	Hunger bar replenishes by +10%; item count decreases; zero waste prop spawned.

Section 1: Problem, Purpose, and Objectives

Questions 1–10

Foundational rationale, legal framework (RA 9003), target audience alignment, and research objectives.

Q1. [Problem, Purpo] EASY

What is the primary real-world problem that Eco Quest aims to address?

Eco Quest addresses the widespread problem of improper solid waste segregation and littering among elementary pupils in the Philippines, as highlighted by Republic Act No. 9003. While many pupils possess theoretical knowledge about waste, traditional awareness campaigns often fail to translate into consistent daily sorting habits. The game bridges this gap by providing an experiential 3D simulation where children actively practice proper waste classification.

> 💡 **Delivery Tip:** Begin by mentioning RA 9003 and state that awareness exists but habits are missing.

Q2. [Problem, Purpo] MEDIUM

Why did your team develop a 3D digital game rather than traditional instructional posters or lectures?

Traditional lectures and posters offer passive learning where students merely read or listen, resulting in low knowledge retention. Research in Game-Based Learning demonstrates that interactive simulations engage learners through active decision-making, immediate visual feedback, and positive reinforcement. In Eco Quest, children physically pick up and throw waste items into color-coded bins, turning abstract concepts into practical muscle memory.

> 💡 **Delivery Tip:** Cite Experiential Learning theory—active practice yields higher retention than passive reading.

Q3. [Problem, Purpo] HARD

Specific Objective 3 promises to 'improve environmental behavior,' but your limitations admit that behavior change cannot be confirmed. How can you claim Objective 3 is met?

We acknowledge this limitation honestly. A 15-minute game evaluation can reliably measure knowledge gain and situational awareness, but confirming permanent real-world behavioral change requires longitudinal observation over months. Our post-quiz scores and 101 student survey responses show that pupils successfully mastered segregation rules in simulated scenarios, which is the necessary cognitive foundation for subsequent habit formation.

> 💡 **Delivery Tip:** Admit the limitation immediately before explaining that simulated mastery builds the foundation for real habits.

Q4. [Problem, Purpo] MEDIUM

Who is the exact target demographic for Eco Quest, and how is the game tailored to their developmental stage?

The target users are primary elementary pupils in Grades 4 to 6 (ages 8 to 12). The game tailors its design to this demographic by using simple sentence structures, readable fonts (Nunito and Fredoka), visual icon keycaps for controls, and bilingual Filipino terms like (Nabubulok) and (Hindi Nabubulok) to support early bilingual readers without language fatigue.

> 💡 **Delivery Tip:** Reference your specific typography (Nunito/Fredoka) and bilingual Tagalog labels.

Q5. [Problem, Purpo] EASY

Why is Republic Act No. 9003 central to your project, and how is it reflected in gameplay?

RA 9003, the Ecological Solid Waste Management Act of 2000, establishes the legal mandate for source segregation and ecological waste management in the Philippines. Eco Quest directly models RA 9003 by utilizing color-coded receptacles (Green for Biodegradable, Blue for Recyclable, and Yellow for Non-Biodegradable) and teaching children the legal categories enforced in local barangays.

> 💡 **Delivery Tip:** Connect the in-game bin colors directly to standard Philippine ecological waste guidelines.

Q6. [Problem, Purpo] HARD

How does the game teach the 'consequences of littering' when dropping trash on the ground incurs zero coin penalty?

Our gameplay separates sorting mistakes from street littering: sorting errors at bins deduct 10 coins on the third strike to penalize classification mistakes, whereas street littering is taught socially through the LitterBug pedestrian mechanic. When a citizen litters, the player actively intervenes, educates the character through dialogue, and restores park cleanliness, reinforcing social accountability rather than punitive game-overs.

> 💡 **Delivery Tip:** Highlight that street littering is modeled as a social intervention rather than an artificial wallet penalty.

Q7. [Problem, Purpo] MEDIUM

What is the role of the 3Rs (Reduce, Reuse, Recycle) in Eco Quest, and where are they taught?

The 3Rs form the core pedagogical framework of the game. Recyclable materials such as paper, cans, and bottles are collected in World 1 and World 2, while educational popups explain how reusing and recycling save energy and forest habitats. In our beta testing, 92.1% of surveyed pupils correctly defined 3Rs, demonstrating effective retention of this principle.

> 💡 **Delivery Tip:** Cite the 92.1% survey score from your Beta Testing results to validate this claim.

Q8. [Problem, Purpo] EASY

What makes Eco Quest distinct from other educational environmental games already available?

Unlike 2D sorting minigames, Eco Quest provides a fully realized 3D dual-world simulation featuring a guided Forest tutorial and an open City Hub with autonomous traffic, pedestrian crowds, and dynamic quest lines. Furthermore, it incorporates localized Philippine context, bilingual Tagalog subtitles, and immediate comparative learning gain assessments.

> 💡 **Delivery Tip:** Emphasize the dual-world structure (Forest tutorial + City Hub) and localized Tagalog terms.

Q9. [Problem, Purpo] HARD

If a public elementary school only has low-end laboratory computers, can Eco Quest run smoothly?

Yes. Eco Quest is built using Unity's Universal Render Pipeline (URP), which is optimized for resource efficiency across varied hardware. Our Settings menu provides Low, Medium, High, and Ultra quality presets, full-screen resolution scaling down to 1024x768, and an FPS limiter, allowing the game to maintain stable 30+ FPS performance even on integrated Intel HD graphics.

> 💡 **Delivery Tip:** Mention your URP engine architecture and the 4-tier graphics quality presets.

Q10. [Problem, Purpo] MEDIUM

Can each team member state their technical role and major contribution to Eco Quest?

Bryan Duarte led game architecture, C# scripting for quest flows, physics pickup, and UI orchestration. Marjorie Pamonag managed 3D scene assembly, lighting, and environmental asset integration. Marianne Rouze Pillado designed the 2D UI elements, quiz data, and dialogue copy. Joanna Niña Tagaca conducted requirements documentation, ISO 25010 evaluations, and beta testing surveys.

> 💡 **Delivery Tip:** Have each student speak their assigned two-sentence part naturally without looking at notes.

Section 2: Scope, Limitations, and Delimitations

Questions 11–20

Project boundaries, excluded platforms, manuscript discrepancies, and intentional design trade-offs.

Q11. [Scope, Limitat] EASY

What is included in the current functional scope of Eco Quest?

The scope covers 3D first-person navigation, a 5-step tutorial, NPC dialogues with camera blending, physics raycast waste collection, a 3-bin segregation quest, pre- and post-educational quizzes, an open City Hub with Ranger Robin's scavenger quest, an Eco Shop with consumables, LitterBug NPC persuasion, and persistent audio/graphics settings.

> 💡 **Delivery Tip:** Summarize the 8 main modules cleanly in one sentence.

Q12. [Scope, Limitat] HARD

Your scope mentions only biodegradable and non-biodegradable waste, but your DFD has 3 categories and the City has 4. Why does this contradiction exist?

This is an acknowledged documentation mismatch resulting from iterative development. The earliest concept focused strictly on bio and non-bio, but during implementation we added recyclable waste to align with standard Philippine 3-bin systems, and added hazardous e-waste in the City database for educational depth. We have prepared formal manuscript corrections to unify all sections to three core categories plus hazardous items.

> 💡 **Delivery Tip:** Acknowledge the version mismatch immediately and offer the prepared manuscript correction.

Q13. [Scope, Limitat] MEDIUM

Why is Eco Quest developed exclusively for Windows desktop instead of mobile devices like Android?

Eco Quest was designed specifically for elementary school computer laboratories, where desktop computers provide standardized hardware, keyboard-mouse inputs, and supervised class usage. Focusing on Windows desktop allowed us to optimize high-fidelity 3D physics, Cinemachine camera blends, and autonomous city traffic without mobile battery or thermal throttling constraints.

> 💡 **Delivery Tip:** Emphasize school computer laboratory deployment as the strategic target.

Q14. [Scope, Limitat] EASY

Why does the game operate without user accounts, student login, or cloud synchronization?

Omitting student logins ensures seamless classroom deployment where multiple students share the same laboratory PC without password management hurdles or session delays. Furthermore, it guarantees 100% compliance with child data privacy laws (COPPA and RA 10173) by ensuring zero personal student information is ever transmitted or stored online.

> 💡 **Delivery Tip:** Frame no-login as an intentional privacy safeguard and zero-friction school setup feature.

Q15. [Scope, Limitat] HARD

Without a teacher dashboard or score export feature, how can educators evaluate student learning outcomes?

We recognize that a teacher dashboard would streamline classroom grading. In the current version, student learning gains are displayed directly on the end-of-quest comparative scorecard, allowing teachers to visually inspect scores in the classroom. In Version 2.0, we plan to implement a local CSV/JSON export module in Application.persistentDataPath for administrative analysis.

> 💡 **Delivery Tip:** Highlight the visible on-screen learning gain scorecard as the current evaluation mechanism.

Q16. [Scope, Limitat] MEDIUM

Why is the Beach scene present in the project folder but omitted from the active Build Settings?

The Beach scene was constructed as a World 3 prototype to explore coastal marine littering scenarios, inspired by the Clean My Beach literature. However, to guarantee optimal performance, rigorous testing, and polish within our capstone timeline, we scoped the official release strictly to World 1 (Forest) and World 2 (City), leaving Beach unlinked for future expansion.

> 💡 **Delivery Tip:** Explain that scoping to Forest and City ensured high polish rather than rushing an unfinished third world.

Q17. [Scope, Limitat] HARD

What prevents a student from bypassing the Daily Quest cooldown by simply advancing their Windows system clock?

Currently, cooldowns are calculated using DateTime.UtcNow on the local system, which means advancing the PC clock can reset the timer. Because Eco Quest is an offline game without server time verification, this is an inherent platform limitation. However, since the goal is educational practice, replaying quests through clock adjustments actually provides additional learning repetitions without harming data integrity.

> 💡 **Delivery Tip:** Acknowledge the offline limitation honestly, then point out that replaying quests still promotes educational practice.

Q18. [Scope, Limitat] MEDIUM

Why are there no in-game level editors or customizable quiz question tools for teachers?

Eco Quest is scoped as a standardized educational curriculum game where trash databases and quiz rationales are scientifically curated to match DepEd elementary competencies. Allowing unverified question editing could introduce incorrect segregation rules or factual inaccuracies. Fixed ScriptableObject databases ensure rigorous educational quality control.

> 💡 **Delivery Tip:** Cite curriculum standardization and educational quality control as the primary reason.

Q19. [Scope, Limitat] EASY

What exact settings can the player adjust in the Settings menu, and do they persist across reboots?

Players can adjust Master Volume, Music Volume, SFX Volume, Mute All, Graphics Quality (Low to Ultra), Fullscreen toggle, Resolution, VSync, FPS Limit (30/60/120/Unlimited), Mouse Sensitivity, Tutorial Hints toggle, and Text Size (Normal vs Large +20%). All settings are saved in PlayerPrefs and persist across game reboots.

> 💡 **Delivery Tip:** List volume sliders, quality dropdowns, and the text size accessibility toggle.

Q20. [Scope, Limitat] HARD

Your limitations claim the game has no accessibility features, yet you implemented Text Size scaling and subtitles. Why does the paper deny features that exist?

The limitation statement was written during the early proposal phase before our UI polish sprint. During development, we recognized that younger pupils needed larger text, so we implemented a +20% font scaling option in SettingsManager.cs alongside high-contrast dialogue subtitle panels. The manuscript limitation text needs an update to accurately credit these implemented accessibility enhancements.

> 💡 **Delivery Tip:** Admit that the proposal limitation was written early and was surpassed during game development.

Section 3: Review of Related Literature and Studies

Questions 21–30

Theoretical grounding, foreign and local research comparisons, citations integrity, and identified gaps.

Q21. [Review of Rela] EASY

What are the primary local and foreign studies that directly influenced the design of Eco Quest?

Foreign studies include 'Cleantopia' by Simanjuntak et al. (2024) for waste popup feedback, 'Trash Tracking Game' by Halida et al. (2024) for early childhood sorting, and Raditya & Pratama (2025) for Octalysis gamification. Local studies include Biol et al. (2025) on Grade 6 pupil habits, Agpuon (2026) on localized Philippine game learning, and Bueta et al. (2023) on circular economy challenges.

> 💡 **Delivery Tip:** Name at least two foreign and two local studies with their publication years.

Q22. [Review of Rela] MEDIUM

How did the foreign study 'Cleantopia' by Simanjuntak et al. (2024) guide your implementation?

Cleantopia evaluated an educational game with Grade 4 students using the Multimedia Development Life Cycle (MDLC), showing that interactive popups before disposal increased sorting awareness. Eco Quest adopted this insight by displaying educational hint popups and fun facts upon waste collection, ensuring students understand why an item belongs to a specific stream before discarding it.

> 💡 **Delivery Tip:** Connect Cleantopia's popup concept to Eco Quest's WrongTrashPopup educational banner.

Q23. [Review of Rela] HARD

Why are 12 of the in-text citations in your literature review completely missing from the References list?

This was an administrative oversight during document assembly where citations added during the literature review drafting were not synchronized with the final references section. We have thoroughly audited all 12 missing entries—including World Bank (2020), Manglicmot et al. (2024), and Halida et al. (2024)—and prepared a complete, APA 7th edition formatted replacement reference list.

> 💡 **Delivery Tip:** Admit the assembly oversight immediately and present the prepared corrected APA reference list.

Q24. [Review of Rela] MEDIUM

What did Qu et al. (2023) discover regarding student attitudes versus actual waste separation practices?

Qu et al. discovered that while students generally possess positive attitudes and high theoretical awareness regarding waste management, their real-world separation practices remain low due to lack of immediate incentives and convenient facilities. Eco Quest addresses this specific gap by linking virtual incentives (Eco-Coins, shop unlocks) to practical simulation tasks, bridging awareness and active execution.

> 💡 **Delivery Tip:** Emphasize that Qu et al. proved awareness alone is insufficient without incentive mechanisms.

Q25. [Review of Rela] HARD

Your reference list contains unformatted raw file names like 'ISWA_Sustainable-Waste-Management_Guide_Digital_.pdf'. Why was this included?

During manuscript compilation, two source PDF file names were accidentally pasted into the bibliography without bibliographic conversion. We have corrected these entries to their official institutional citations: the International Solid Waste Association (ISWA) Sustainable Waste Management Guide (2021) and Satria et al. (2021) in the IOP Conference Series.

> 💡 **Delivery Tip:** State the official titles of both documents and present the formal APA citation.

Q26. [Review of Rela] MEDIUM

How does the local study by Agpuon (2026) on Gen Z-IGNOS justify your use of localized Tagalog terms?

Agpuon (2026) demonstrated that localized, culturally contextualized educational games significantly enhance Filipino students' comprehension and engagement compared to purely foreign software. By incorporating parenthetical Filipino terms like (Nabubulok) and (Pwedeng Gamitin Muli), Eco Quest bridges national curriculum guidelines with vernacular terms familiar to young learners.

> 💡 **Delivery Tip:** Cite Agpuon's conclusion that localization improves comprehension among Filipino learners.

Q27. [Review of Rela] HARD

Several studies in your review, such as Hairul et al. (2024), lacked empirical test scores. Why include studies with incomplete data?

We included Hairul et al. not for their empirical statistical findings, but for their game design architecture and requirements analysis, which implemented Unity-based drag-and-drop sorting and quizzes for elementary pupils aged 5 to 9. Their methodological framework provided valuable structural benchmarks for our own system development.

> 💡 **Delivery Tip:** Clarify that the study was cited for its game design methodology rather than empirical results.

Q28. [Review of Rela] EASY

What is the primary synthesis of your literature review, and what research gap does Eco Quest fill?

The literature synthesis establishes that educational games improve environmental awareness, but existing projects are often small, single-screen minigames tested on narrow samples without standardized software quality evaluations. Eco Quest fills this gap by offering a full 3D multi-biome world evaluated rigorously against ISO 25010 functional suitability with 101 elementary pupils.

💡 **Delivery Tip:** Emphasize your 3D multi-biome world and formal ISO 25010 evaluation as your key gap.

Q29. [Review of Rela] MEDIUM

How does the study by Biol et al. (2025) on Grade 6 pupils validate your chosen target audience?

Biol et al. (2025) surveyed elementary pupils in the Philippines and concluded that environmental education delivered during Grade 6 has the highest positive correlation with forming lifelong recycling habits. This validates our design decision to tailor the game's vocabulary, quests, and mechanics specifically to pupils in Grades 4 through 6.

💡 **Delivery Tip:** Mention Biol's finding that elementary school is the formative window for environmental habit formation.

Q30. [Review of Rela] HARD

Your literature review cites Villagómez-Palacios et al. (2026) on Unity Visual Scripting, but your game uses C#. Why cite Visual Scripting if you wrote C#?

We cited Villagómez-Palacios to establish the pedagogical efficacy of the Unity engine in educational software development. While that paper analyzed Visual Scripting for novice learners, our team leveraged industry-standard C# scripting to achieve greater performance optimization, custom event decoupling, and physics raycast control. We will clarify this distinction in the final revision.

💡 **Delivery Tip:** Explain that Unity was the core subject of the citation, while C# was chosen for performance and scalability.

Section 4: Game Design and Educational Value

Questions 31–40

Mechanics, core gameplay loops, player progression, pedagogical feedback, and reward-penalty structures.

Q31. [Game Design an] EASY

Can you walk us through the complete gameplay loop of World 1 (Forest Tutorial)?

The player enters Forest Scene, completes a 5-step movement tutorial (Look, WASD, Sprint, Map, Cursor), and speaks with the Guardian NPC. After completing a 6-question Pre-Quiz, the player collects and sorts 9 scattered trash items into Green, Blue, and Yellow bins. Upon completing the sort, the player takes the Post-Quiz, views their learning gain scorecard, receives 100-150 coins, and talks to the Explorer to unlock the City.

> 💡 **Delivery Tip:** Walk through the sequence smoothly: Tutorial -> Guardian -> Pre-Quiz -> Sorting -> Post-Quiz -> Gate.

Q32. [Game Design an] MEDIUM

How does the game differentiate between Biodegradable, Recyclable, and Non-Biodegradable items in the Forest Scene?

In the Forest Scene, items are tagged and categorized: organic fruit and food scraps (banana peel, fish bone) are Biodegradable; clean cardboard and tin cans are Recyclable; single-use plastic cups are Non-Biodegradable. Bins use color-coding (Green, Blue, Yellow) and bilingual labels, and dropping an item into the wrong bin immediately displays an educational hint explaining why.

> 💡 **Delivery Tip:** Name the 3 bin colors and give at least one specific item example for each.

Q33. [Game Design an] HARD

What exact feedback and penalty occur when a player deposits an item into the wrong bin?

When an item enters the wrong bin, the bin plays an error buzzer, ejects the item back out with a physics impulse, and triggers the WrongTrashPopup banner displaying '💡 Hint: Put this in the [Correct Color] Bin! • Mistake X/3'. On the first two mistakes, no coins are lost; on the third consecutive mistake, 10 Eco-Coins are deducted (clamped at 0) and the counter resets, teaching through guidance rather than punitive failure.

> 💡 **Delivery Tip:** Detail the 3-mistake rule: warning on 1 and 2, -10 coins on 3, item always physically ejected.

Q34. [Game Design an] EASY

What is the purpose of the Pre-Quiz and Post-Quiz, and how is the learning gain calculated?

The Pre-Quiz establishes the pupil's baseline knowledge before playing, while the Post-Quiz measures knowledge acquired after completing the sorting activities. Learning gain is calculated as Post-Quiz Score minus Pre-Quiz Score, and the result is presented on an animated scorecard that visually demonstrates how much the student learned during gameplay.

> 💡 **Delivery Tip:** State the formula: Learning Gain = Post-Quiz Score - Pre-Quiz Score.

Q35. [Game Design an] MEDIUM

Why are the Pre-Quiz and Post-Quiz question pools identical, and how do you prevent simple memorization bias?

Using identical questions is standard educational testing methodology to ensure valid pre- and post-intervention comparison. To minimize simple position recall, QuizManager.cs applies a Fisher-Yates shuffle to randomize both question order and answer choices. Furthermore, the Pre-Quiz hides all explanations, ensuring pupils cannot simply memorize correct answers prior to playing.

> 💡 **Delivery Tip:** Mention the Fisher-Yates shuffle and the fact that Pre-Quiz provides zero answer feedback.

Q36. [Game Design an] HARD

In the City Scene, what is the role of Ranger Robin, and how does the Trash Detective quest differ from Forest sorting?

While the Forest focuses on static segregation rules, Ranger Robin's Trash Detective quest in the City simulates an urban park cleanup. The player must find 10 randomly spawned trash items across 5 city districts within a 4:00 minute timer, utilizing an in-game compass radar [F]. It reinforces real-world environmental identification under active exploration conditions.

> 💡 **Delivery Tip:** Emphasize that City teaches active trash spotting and urban cleanup, complementing Forest segregation.

Q37. [Game Design an] MEDIUM

What is the LitterBug mechanic, and why is it designed as an NPC persuasion interaction rather than combat?

LitterBugs are ordinary civilian pedestrians walking the city streets who occasionally toss trash props when the player is nearby. The player approaches within 2.4 meters and presses [E] to engage in persuasive dialogue, prompting the citizen to apologize, stop littering, and clean up. This teaches restorative civic intervention rather than violent or adversarial conflict.

> 💡 **Delivery Tip:** Explain that peaceful persuasion teaches real-world civic responsibility suitable for children.

Q38. [Game Design an] CONFIRM HARD

In the Eco Shop, players purchase food and drinks. Why does consuming these items not generate trash wrappers or bottles?

CONFIRM: In the current build, consuming purchased food directly restores hunger points without instantiating empty wrapper props. This was an intentional scope boundary to prevent the backpack inventory from overflowing with useless trash props. However, as an educational game, adding consumable waste that must be disposed of is an excellent suggestion that we have prioritized for Version 2.0.

> 💡 **Delivery Tip:** Tag CONFIRM, admit the design trade-off, and present it as an active enhancement for Version 2.0.

Q39. [Game Design an] EASY

How does the Objective Marker system assist young players who become disoriented?

ObjectiveMarkerService renders a 3D animated bouncing gold arrow directly above active objective targets, such as the Guardian NPC, bins, or Ranger Robin. When the target is off-screen, a screen-edge pointer appears with real-time distance telemetry in meters, ensuring pupils always know which direction to travel.

> 💡 **Delivery Tip:** Mention the bouncing 3D gold arrow and the screen-edge meter distance telemetry.

Q40. [Game Design an] HARD

Why does dropping trash on the ground have zero penalty, while sorting incorrectly incurs a 10-coin deduction?

This design choice prevents young children from being punished while actively managing items. During gameplay, pupils frequently drop items to reorganize their hands or test physics. Penalizing ground drops would frustrate children and discourage experimentation. Instead, penalties are reserved exclusively for depositing waste into an incorrect bin, reinforcing deliberate sorting accuracy.

> 💡 **Delivery Tip:** Explain that ground drops are often accidental or organizational, while bin deposits represent conscious choices.

Section 5: Technical Implementation and Architecture

Questions 41–50

Unity engine internals, C# scripts, physics casts, singleton architecture, save pipelines, and crash fallbacks.

Q41. [Technical Impl] EASY

What is the underlying technical stack and architectural foundation of Eco Quest?

Eco Quest is built on Unity 6000.3.11f1 (URP 17.6) using C# (.NET Standard 2.1). The architecture utilizes singleton service managers (UIOrchestrator, ModalManager, SettingsManager) combined with ScriptableObject data containers (QuizData, CityTrashDatabase) to decouple game logic from scene assets and UI view layers.

> 💡 **Delivery Tip:** Cite Unity 6, URP 17.6, C#, singletons, and ScriptableObjects.

Q42. [Technical Impl] HARD

Can you explain the exact physics and raycast mechanism used for the drag-and-drop trash pickup system?

DragAndDrop.cs casts a ViewportPointToRay from the camera center (0.5, 0.5) with a SphereCast fallback of 0.12m radius up to 5.0m range against the TrashItem layer. When grabbed, the object's Rigidbody is made kinematic, colliders are set to non-solid triggers, and its transform is lerped at speed 25 to the camera's hold socket. Pressing [G] restores colliders, applies 3.5 impulse force, and wakes the Rigidbody.

> 💡 **Delivery Tip:** Explain the 5m SphereCast, kinematic socket lerp at speed 25, and [G] impulse throw.

Q43. [Technical Impl] MEDIUM

How are UI modals, dialogue boxes, and HUD visibility decoupled across scenes?

UIOrchestrator.cs coordinates with ModalManager.cs through state machine profiles. When a dialogue or quiz opens, ModalManager requests the modal, fades out the gameplay HUD via HudVisibilityService, and manages cursor unlock state. When closed, it safely pops the modal stack and restores the HUD, preventing overlapping UI panels and input lockouts.

> 💡 **Delivery Tip:** Mention UIOrchestrator, ModalManager, and HudVisibilityService working together.

Q44. [Technical Impl] HARD

How does QuestManager.cs validate whether a trash item matches a bin, and what is the fallback if an item is untagged?

QuestManager.IsItemMatchingCategory first compares item.CompareTag(acceptedTag). If tags are empty or unmatched, it executes a lowercase string fallback checking whether the item's GameObject name contains category keywords like 'bio', 'apple', 'banana', 'food', or 'recyc', 'can', 'bottle', 'paper'. This dual-layer validation ensures third-party mesh prefabs still sort correctly.

> 💡 **Delivery Tip:** Explain the two layers: primary tag comparison, followed by substring keyword matching on GameObject name.

Q45. [Technical Impl] EASY

What is the purpose of Cinemachine in your project, and where is it utilized?

Cinemachine 3.1.6 is utilized for smooth, cinematic camera blending. During NPC interactions with the Guardian or Ranger Robin, Cinemachine dynamically blends the camera from the first-person perspective into a framed medium shot of the NPC, and smoothly returns to first-person view once dialogue concludes.

> 💡 **Delivery Tip:** Highlight the smooth camera transition from first-person into framed NPC dialogue.

Q46. [Technical Impl] MEDIUM

How is the autonomous vehicle traffic system implemented in the City Scene without causing vehicle collisions or gridlock?

CityTrafficManager.cs coordinates vehicle paths across 5 districts using waypoint node networks. Each TrafficVehicleController projects forward raycasts to detect preceding vehicles, pedestrians, or obstacles, dynamically adjusting speed and applying brake deceleration. Turn arc reservations at intersections prevent turning vehicles from overlapping or deadlocking.

> 💡 **Delivery Tip:** Mention waypoint node paths, forward obstacle raycast braking, and intersection turn reservations.

Q47. [Technical Impl] HARD

How does the save system operate, where is data physically saved on Windows, and what happens if a save key is missing?

Data is saved via Unity's PlayerPrefs in the Windows Registry under HKCU\Software\DefaultCompany\EcoQuest. It stores keys for pre/post quiz scores, coins, unlocked quests, inventory strings, and settings. If a key is missing or unreadable, PlayerPrefs.GetInt/GetString safely returns default fallback values (0 coins, empty inventory), preventing game crashes.

> 💡 **Delivery Tip:** State the physical Windows Registry path and emphasize graceful fallback defaults without crashes.

Q48. [Technical Impl] MEDIUM

How does the dynamic map and minimap system track player heading, quest icons, and NPC positions?

MinimapController.cs manages both the corner HUD minimap and full-screen map [M]. It translates the player's 3D XZ coordinates into normalized UV coordinates on the map image. Tracked entities (player arrow, NPCs, bins) register MinimapIcon components that dynamically update position and rotation on the radar canvas every frame.

> 💡 **Delivery Tip:** Explain the 3D XZ to 2D UV coordinate translation and MinimapIcon tracking components.

Q49. [Technical Impl] HARD

Recently, a bug prevented the wrong-trash notification popup from appearing. What was the root cause and fix?

In Forest Scene.unity, the WrongTrashPopup GameObject was saved inactive (m_IsActive: 0). In Unity, Awake() never executes on inactive GameObjects, leaving WrongTrashPopup.Instance as null, so QuestManager silently skipped calling Show(). We fixed it by updating the Instance getter to auto-resolve using FindObjectsInactive.Include, auto-activating the GameObject, and registering an AfterSceneLoad bootstrap.

> 💡 **Delivery Tip:** Explain that Awake() never runs on inactive objects, and that FindObjectsInactive.Include resolved it.

Q50. [Technical Impl] HARD

If the game crashes, freezes, or encounters an unexpected Unity error during our live defense demo, what is your fallback protocol?

Our defense protocol has a two-step safeguard: first, our settings and save system can be restored instantly via our QA reset menu or restarting the build. Second, we have prepared a full, unedited high-definition video recording of end-to-end gameplay demonstrating all quests, quizzes, and city interactions ready on our defense laptop to present immediately without stalling the panel.

> 💡 **Delivery Tip:** Deliver this confidently: state the instant reboot protocol and point to your backup gameplay video.

Section 6: Methodology and Theoretical Framework

Questions 51–60

DevOps lifecycle justification, GBL principles, Behavioral Reinforcement, and timeline analysis.

Q51. [Methodology an] HARD

Why did you choose DevOps as your development methodology for an offline desktop game without cloud infrastructure?

We adopted DevOps principles adapted for game engineering: continuous integration of modular gameplay systems, automated regression test runners (such as UIPhaseMatrixTestRunner), and rapid feedback loops between scene designers and scripting. While we do not operate cloud servers, the iterative build-test-monitor cycle ensured high software stability, zero Play Mode errors, and consistent multi-scene builds.

💡 **Delivery Tip:** Explain that you applied DevOps principles (automated testing, rapid iteration) adapted for game stability.

Q52. [Methodology an] MEDIUM

How was the testing and monitoring phase of DevOps practically carried out during your development cycles?

We developed specialized in-editor test suites, including automated UI stress tests that execute rapid modal cycles, verify camera blending, and inspect inventory slot states. Monitoring was performed by parsing Unity's Editor.log for memory allocations and script compilation warnings, allowing us to eliminate memory leaks and NullReferenceExceptions before beta testing.

💡 **Delivery Tip:** Mention your automated UIPhaseMatrix test suite and Editor.log profiling.

Q53. [Methodology an] EASY

How does Game-Based Learning (GBL) guide the design of Eco Quest?

Game-Based Learning integrates educational objectives directly into game mechanics rather than treating learning as an external quiz. In Eco Quest, players do not merely read about waste categories; they explore environments, physically classify 3D debris, experience immediate visual feedback, and earn in-game rewards, making learning active, engaging, and memorable.

💡 **Delivery Tip:** State that learning is embedded directly inside core game actions, not bolted on as text.

Q54. [Methodology an] MEDIUM

How does Behavioral Reinforcement Theory operate within your coin reward and mistake penalty systems?

B.F. Skinner's Operant Conditioning is applied via positive reinforcement and negative consequences. Correct segregation yields positive audio chimes, celebratory visual effects, and coin rewards. Incorrect segregation triggers an informative warning and an ejection impulse, followed by a 10-coin penalty on repeated errors. This teaches students that environmental actions have direct tangible consequences.

💡 **Delivery Tip:** Cite B.F. Skinner and Operant Conditioning: positive reward for correct, corrective penalty for error.

Q55. [Methodology an] MEDIUM

In what way does Experiential Learning theory justify a 3D simulation over a 2D web quiz?

David Kolb's Experiential Learning Cycle posits that knowledge is created through the transformation of concrete experience into reflective observation. A 3D simulation provides spatial presence where pupils observe realistic street scenes, interact with pedestrians, and manipulate physical objects, offering a far closer approximation of real-world physical waste sorting than flat 2D multiple-choice forms.

💡 **Delivery Tip:** Cite Kolb's Experiential Learning Cycle: concrete experience leads to reflective observation.

Q56. [Methodology an] HARD

Your Gantt chart includes Traffic System and Shop System, but they are missing from your paper's Scope. Why this discrepancy?

Our Gantt chart reflects the true engineering schedule from February to November 2026, during which our team implemented the autonomous vehicle routing and Handyman Eco Shop to enrich the City Hub. The Scope section in the proposal draft was written before these modules were completed. We have prepared an amendment formally incorporating these two systems into the Scope.

💡 **Delivery Tip:** Acknowledge that the Gantt chart reflects true development progress and offer the Scope amendment.

Q57. [Methodology an] EASY

What code editor and integrated development environment were utilized alongside Unity 6?

The team utilized Visual Studio Code configured with the official Unity extension and C# Dev Kit. This environment provided intelligent code completion, semantic linting, and integrated debugging across our project scripts.

💡 **Delivery Tip:** Name Visual Studio Code with C# Dev Kit and Unity tools.

Q58. [Methodology an] MEDIUM

How did your team manage Git version control and branch merging during development?

We maintained our repository on GitHub, utilizing feature branches for distinct subsystems such as city traffic, dialogue UI polish, and inventory mechanics. Commits followed conventional commit syntax (feat, fix, chore), and checkpoint commits were made before merging major asset updates to ensure regression-free rollbacks.

💡 **Delivery Tip:** Mention feature branches, conventional commits, and checkpoint backups.

Q59. [Methodology an] HARD

How did you ensure child-friendly UX standards when translating theoretical concepts into game mechanics?

We adhered to child-computer interaction guidelines: eliminating punitive game-over screens, providing visual keycaps (showing literal [E] and [Q] keys), using high-contrast color palettes (WCAG AAA compliant dark green), and ensuring that text display speeds allow Grade 4 pupils ample time to read without feeling rushed.

> 💡 **Delivery Tip:** Cite keycap visuals, no instant game-overs, and WCAG high-contrast readability.

Q60. [Methodology an] MEDIUM

How were development responsibilities divided across your 4-member capstone team?

Responsibilities were allocated by specialization: Bryan Duarte served as lead programmer (game logic, physics, traffic, UI architecture); Marjorie Pamonag acted as level and environment designer (scene construction, lighting, URP assets); Marianne Rouze Pillado managed 2D assets, copy, and quiz curation; and Joanna Niña Tagaca led documentation, ISO evaluation, and beta testing execution.

> 💡 **Delivery Tip:** State clearly that every member had a distinct, ownership-driven role.

Section 7: Requirements and System Design (DFD, ISO 25010)

Questions 61–70

Data Flow Diagrams (Level 0, 1, 2), ISO 25010 quality claims, UML architecture, and schema verification.

Q61. [Requirements a] HARD

Can you explain Data Flow Diagram (DFD) Level 0, including its external entities, data flows, and data stores?

In DFD Level 0, the single external entity is User/Player, who provides inputs (WASD, Mouse, Raycast E, Grab/Throw LMB, Quiz answers, Shop clicks) to Process 0.0 (Eco Quest Game Engine). The engine outputs the 3D Viewport Render, Survival HUD, Scorecards, Audio FX, and Error Popups. It interfaces with two data stores: D1 (Player Data), which is read/written for saves and coins, and D2 (Asset Database), which is read-only for quiz and trash definitions.

> 💡 **Delivery Tip:** Explain the entity (Player), process 0.0 (Game Engine), and the two data stores (D1 Player, D2 Asset Database).

Q62. [Requirements a] HARD

Can you walk us through DFD Level 1, explaining how process 3.0 connects to process 4.0 and data store D1?

DFD Level 1 breaks the engine into six processes: 1.0 Menu/Settings, 2.0 Movement/Tutorial, 3.0 Waste Sorting, 4.0 Quiz Evaluation, 5.0 Eco Shop, and 6.0 Show HUD. Process 3.0 reads item tags and bin data, recording deposit counters and mistakes to D1. Upon completing 3.0, it triggers Process 4.0, which reads questions from D3, evaluates answers, calculates learning gains, and saves post-test scores and gold rewards back to D1.

> 💡 **Delivery Tip:** Trace the handoff: Process 3.0 finishes sorting, updates D1, and triggers Process 4.0 for quiz evaluation.

Q63. [Requirements a] HARD

Explain the 7-step flow of DFD Level 2 for the Forest sorting quest, specifically how process 3.4 validates bin triggers.

DFD Level 2 details the sorting loop: 3.1 sets the quest target (3 Bio, 3 Recyc, 3 Non-Bio); 3.2 executes Raycast & Grab; 3.3 handles Carry & Throw via hold socket; 3.4 compares item tag vs bin accepted tag. If matching, 3.5 deposits the item, stacks the visual layer, and updates count. If mismatched, 3.6 ejects the item, shows an alert, and increments mistakes. When all targets are met, 3.7 awards gold and initiates the post-quiz.

> 💡 **Delivery Tip:** Walk through 3.1 to 3.7 in sequence, highlighting step 3.4's branching on Tag Match vs Mismatch.

Q64. [Requirements a] MEDIUM

In DFD Level 0, data store D2 is labeled 'Asser Database'. What does this represent, and why did this typo occur?

This is a typographical error on the diagram graphic where 'Asset Database' was misspelled as 'Asser Database'. D2 represents the ScriptableObject databases, specifically QuizData.asset and CityTrashDatabase.asset, which store question rationales, trash item metadata, and educational notes. We have corrected this typographical error in our updated diagram sheet.

> 💡 **Delivery Tip:** Acknowledge the typo plainly, clarify that it means Asset Database, and cite QuizData.asset.

Q65. [Requirements a] HARD

In your ISO 25010 table under Confidentiality, you claim the system restricts access to game data. How is this true if PlayerPrefs is unencrypted in the Registry?

We acknowledge this as an overclaim in the manuscript. PlayerPrefs writes plaintext values to the Windows Registry, meaning an advanced user with regedit can view or modify coins and scores. However, in our target school laboratory context with Grade 4 to 6 pupils operating standard non-administrator user accounts, Windows permissions prevent casual tampering. In Version 2.0, we plan to implement AES encryption on save keys.

> 💡 **Delivery Tip:** Admit the plaintext PlayerPrefs reality, cite non-admin lab accounts as the practical defense, and propose AES.

Q66. [Requirements a] HARD

Under Authenticity in ISO 25010, you claim the game ensures assets originate from the official app. Does your code use digital signatures or hashing?

No, the current build does not implement cryptographic hashing or digital asset signatures; assets are compiled directly into Unity's proprietary .assets and .sharedAssets binary bundles. While standard pupils cannot easily alter bundled assets, the manuscript statement overstates this as a dedicated security feature. We will revise the Authenticity claim to accurately reflect compiled binary bundling rather than cryptographic verification.

> 💡 **Delivery Tip:** Clarify that compiled Unity binary bundles provide standard packaging, not cryptographic authenticity.

Q67. [Requirements a] MEDIUM

How does your game satisfy ISO 25010 Functional Completeness and Functional Correctness?

Eco Quest satisfies Functional Completeness by implementing all specified user journey components: tutorial, exploration, waste sorting, quizzes, shop, and settings. It satisfies Functional Correctness by ensuring that all sorting actions trigger accurate classification, correct coin rewards, proper mistake penalties, and exact learning gain score computations as verified in our automated QA suites.

> 💡 **Delivery Tip:** Address completeness (all modules exist) and correctness (sorting and scores work accurately).

Q68. [Requirements a] EASY

What measures were taken under ISO 25010 Usability to ensure elementary pupils can easily operate the game?

Usability was achieved through intuitive first-person controls, a step-by-step movement tutorial, visual on-screen keycap icons ([E], [Q], [G], [M]), automated 3D objective arrows, persistent dialogue subtitle boxes, and an uncluttered HUD displaying clear progress bars.

> 💡 **Delivery Tip:** List the tutorial, keycap icons, 3D arrows, subtitles, and progress bars.

Q69. [Requirements a] MEDIUM

How does the system handle Fault Tolerance and Recoverability if an unexpected script exception occurs?

Crucial systems implement defensive programming: DragAndDrop.cs features automatic desync recovery and settle failsafes that prevent trash items from getting stuck inside geometry; ModalManager utilizes watchdog routines that auto-restore the HUD if a transition hangs; and PlayerPrefs safely returns default zeros if save keys are missing.

> 💡 **Delivery Tip:** Cite the trash settle failsafe, ModalManager watchdog, and PlayerPrefs default fallbacks.

Q70. [Requirements a] HARD

Your Use Case diagram lists an action labeled 'execute agonic score load.' What does this mean?

This is a transcription typo on the diagram graphic resulting from OCR/text conversion; the intended technical phrase is 'execute async scene load' via Unity's SceneManager.LoadSceneAsync. It represents the asynchronous background loading of the City Scene while the interactive loading screen displays educational trivia.

> 💡 **Delivery Tip:** Clarify the typo immediately: it means 'async scene load' using SceneManager.LoadSceneAsync.

Section 8: Testing and Evaluation

Questions 71–80

Beta testing methodology, 101 student sample, knowledge assessment analysis, survey results, and manuscript gaps.

Q71. [Testing and Ev] EASY

How many respondents participated in your beta testing evaluation, and what was their demographic profile?

A total of 101 elementary pupils participated: 34.7% were Grade 4, 33.7% were Grade 5, and 31.7% were Grade 6, spanning ages 8 to 12. This balanced distribution across all three target elementary grade levels ensured that our evaluation represented the intended primary demographic.

> 💡 **Delivery Tip:** State the numbers: 101 pupils, 34.7% Gr4, 33.7% Gr5, 31.7% Gr6, ages 8-12.

Q72. [Testing and Ev] HARD

In your Beta Testing results for Question 1, 42.6% answered Plastic Bottle as biodegradable while only 41.6% chose Banana Peel. Why did the plurality fail this question?

This is the single most critical finding in our evaluation: 42.6% of pupils misclassified plastic bottles as biodegradable. This highlights a prevalent real-world misconception among children that recyclable plastic is biodegradable because it is 'reusable.' While the manuscript text glossed over this as positive, this result proves precisely why games like Eco Quest are urgently needed, and shows where our tutorial hints must be strengthened.

> 💡 **Delivery Tip:** Own this finding boldly: admit that 42.6% failed, explain the pupil misconception, and show why it proves the game's necessity.

Q73. [Testing and Ev] HARD

Question 2 of your knowledge assessment includes 'Residual' and 'Hazardous' bins. Why test on categories not in the Forest tutorial?

Question 2 evaluated whether students could correctly identify old newspaper as Recyclable (65.3% answered correctly). Options 'Residual' and 'Hazardous' were included as standard real-world distractors matching standard DepEd science terminology, and to preview hazardous items introduced later in the City Trash Detective quest (e.g., batteries, light bulbs).

> 💡 **Delivery Tip:** Explain that 'Residual' and 'Hazardous' served as realistic distractors matching DepEd terminology.

Q74. [Testing and Ev] MEDIUM

What were the overall satisfaction ratings across the 15 evaluation criteria in your UAT survey?

Across all 15 survey criteria, student ratings were overwhelmingly positive, with over 80% selecting 4 (Agree) or 5 (Strongly Agree). Criteria evaluating ease of control, visual design, and educational value received the highest concentration of 5-star ratings, confirming high pupil engagement.

> 💡 **Delivery Tip:** Highlight that over 80% of student ratings fell into the top two agreement categories (4 and 5).

Q75. [Testing and Ev] MEDIUM

What software testing methodologies were executed before conducting beta testing with elementary students?

Before pupil evaluation, we conducted Black-Box functional testing to verify all quest triggers, boundary collision checks to prevent falling through terrain, input bot regression passes to ensure dialogue choices respond to keys, and automated UI stress tests to ensure rapid modal cycling caused zero script exceptions.

> 💡 **Delivery Tip:** Mention Black-Box testing, boundary collision tests, and automated UI stress runs.

Q76. [Testing and Ev] HARD

How did your team ensure the ethical and safe treatment of 101 minor elementary students during beta testing?

Testing was conducted under formal institutional coordination with school administrators and teachers. Surveys were completely anonymous, collecting zero names, student IDs, or photographs. Pupils played in a supervised, comfortable classroom setting with teacher presence, and voluntary participation was reinforced at all times.

> 💡 **Delivery Tip:** Emphasize institutional school coordination, teacher supervision, and complete survey anonymity.

Q77. [Testing and Ev] MEDIUM

What were the most frequent suggestions provided by students in the open-ended feedback section?

Students overwhelmingly asked for more content: additional levels, more city maps, new vehicle types, more NPC quests, and expanded character customization. This feedback confirmed high player immersion, as children wanted to continue exploring beyond the current two worlds.

> 💡 **Delivery Tip:** Highlight that students requested more levels, maps, and quests, proving strong engagement.

Q78. [Testing and Ev] HARD

Your survey evaluated pupils immediately after playing. How can you demonstrate that this knowledge is retained weeks later?

We acknowledge that our cross-sectional study measured immediate learning acquisition rather than longitudinal memory retention. To prove long-term retention in future research, a follow-up assessment administered 30 days post-intervention would be required. However, the high immediate gain demonstrates that the interactive game mechanics successfully conveyed core segregation concepts.

💡 **Delivery Tip:** Admit that immediate post-tests measure acquisition, and propose a 30-day delayed post-test for future research.

Q79. [Testing and Ev] EASY

Which specific game feature received the most praise from student participants?

According to the open-ended responses, the waste collection and drag-and-drop physics sorting was the favorite feature. Students expressed excitement about picking up 3D items, carrying them, and throwing them into matching bins, confirming that experiential physics mechanics are far more engaging than flat 2D buttons.

💡 **Delivery Tip:** Point to the physical drag-and-drop picking and throwing as the favorite pupil mechanic.

Q80. [Testing and Ev] HARD

Why are the tables for Implementation Plan, Resources Needed, Specific Requirements, and Conclusion blank in your paper?

This was an incomplete draft export issue prior to defense submission where terminal appendices and implementation schedules were left in template form. We have finalized all four tables: defining the 5-phase rollout plan, detailing hardware/software deployment requirements, documenting system specifications, and writing a comprehensive research conclusion to complete the manuscript.

💡 **Delivery Tip:** Acknowledge the template export issue honestly and state that completed tables are ready for submission.

Section 9: Ethics, Data Privacy, and Security

Questions 81–90

Child data protection (COPPA, RA 10173), registry security, asset intellectual property, and offline safety.

Q81. [Ethics, Data P] EASY

Does Eco Quest collect, transmit, or store any personal information from the children who play it?

No. Eco Quest collects zero personal identifying information. The game has no student name prompts, no account registration, no student ID input, no microphone or camera access, and no network transmission sockets. It is completely anonymous and offline.

> 💡 **Delivery Tip:** State firmly: zero personal data collected, stored, or transmitted.

Q82. [Ethics, Data P] MEDIUM

How does Eco Quest comply with the Philippine Data Privacy Act of 2012 (RA 10173) and child protection standards?

Eco Quest adheres to the principle of Privacy by Design. Under RA 10173 and COPPA standards, protecting minors requires minimizing data collection to only what is strictly necessary. By collecting zero personal data and storing only anonymous gameplay integers locally, Eco Quest eliminates any risk of student data exposure.

> 💡 **Delivery Tip:** Cite Privacy by Design and data minimization under RA 10173 and COPPA.

Q83. [Ethics, Data P] HARD

If a pupil edits the Windows Registry to grant themselves 999,999 coins, does this pose an academic integrity risk?

In a competitive or graded scenario, client-side registry editing would be a vulnerability. However, Eco Quest is a single-player pedagogical learning tool, not an accredited examination portal. If a pupil explores regedit, it affects only their local client without impacting peers. For school deployments, standard lab user accounts restrict regedit access entirely.

> 💡 **Delivery Tip:** Point out that it's a formative learning tool, not a graded exam, and school PC policies block regedit.

Q84. [Ethics, Data P] MEDIUM

What parental or institutional consent protocols were observed prior to surveying the 101 minor students?

Beta testing was conducted under school administrative authority and teacher supervision during regular class laboratory periods. The evaluation focused strictly on software usability and basic science knowledge, with completely anonymous survey sheets that ensured no student identities could be tracked or publicized.

> 💡 **Delivery Tip:** Mention school administrative approval, teacher supervision, and complete survey anonymity.

Q85. [Ethics, Data P] HARD

How did your team handle intellectual property and licensing for third-party 3D models and audio assets?

All third-party environment packs (Polyperfect Low Poly Pack, Polytope Studio, Advanced People System) were acquired under valid Unity Asset Store Standard EULA licenses permitting commercial and educational software compilation. Audio clips were sourced under Creative Commons Zero (CC0) public domain licenses, ensuring full copyright compliance.

> 💡 **Delivery Tip:** Cite Unity Asset Store Standard EULA for 3D assets and Creative Commons Zero (CC0) for audio.

Q86. [Ethics, Data P] MEDIUM

Are the fonts used in the game (Nunito and Fredoka) legally licensed for educational distribution?

Yes. Both Nunito and Fredoka are open-source typefaces sourced from Google Fonts and licensed under the SIL Open Font License (OFL 1.1). This license explicitly permits bundling, embedding, and distribution within digital applications without licensing fees or royalties.

> 💡 **Delivery Tip:** Name the SIL Open Font License (OFL 1.1) for both Nunito and Fredoka.

Q87. [Ethics, Data P] HARD

If this game were installed across public elementary school computers, how do you ensure the executable is safe from malware?

Eco Quest is built as a self-contained, standalone Windows x86_64 executable without external DLL dependencies, internet downloaders, or shell script hooks. The build can be scanned and verified by standard Windows Defender antimalware suites, and operates within standard restricted user permissions without requiring administrator rights.

> 💡 **Delivery Tip:** State that the build requires no administrator privileges and has no external web downloaders.

Q88. [Ethics, Data P] EASY

Why is operating completely offline an ethical and practical advantage for Philippine public schools?

Many public elementary schools in the Philippines face inconsistent, limited, or completely absent internet connectivity in their computer laboratories. An offline application ensures equitable access for all pupils regardless of institutional internet funding, while eliminating any vulnerability to online web trackers or advertising.

> 💡 **Delivery Tip:** Highlight equitable access for schools without internet, and zero online advertising.

Q89. [Ethics, Data P] MEDIUM

What ethical considerations went into portraying the LitterBug as an ordinary citizen rather than a monster or villain?

We deliberately modeled the LitterBug as an ordinary city pedestrian to reflect real life: littering is committed by everyday citizens due to carelessness, not cartoon villains. Presenting the interaction as a polite educational persuasion teaches children empathy, constructive dialogue, and civic responsibility rather than stigmatization.

> 💡 **Delivery Tip:** Explain that real-world littering is done by ordinary people, so persuasion teaches civic empathy.

Q90. [Ethics, Data P] CONFIRM HARD

When players consume food in the Eco Shop, their hunger is restored but no nutrition lesson is given. Does this cause misconceptions?

CONFIRM: In the current version, the Eco Shop functions primarily as a gameplay stamina-replenishment loop. While items feature healthy options like apples and veggie sandwiches alongside treats, the shop does not teach nutritional food pyramids. In Version 2.0, we plan to incorporate basic nutritional facts and require pupils to segregate the resulting food containers.

> 💡 **Delivery Tip:** Tag CONFIRM, acknowledge the gameplay mechanic, and propose nutritional container sorting for V2.

Section 10: Impact, Sustainability, and Tough Questions

Questions 91–100

Real-world viability, curriculum integration, future roadmap, panel defense strategy, and final closing pitch.

Q91. [Impact, Sustai] HARD

If approved, what is your realistic deployment and rollout strategy for public elementary schools in Manila?

Our rollout strategy follows a 3-phase school laboratory deployment: Phase 1 provides standalone offline USB installer packages to pilot partner schools in Sta. Mesa; Phase 2 trains Grade 4-6 science teachers to integrate the 15-minute game session into their solid waste curriculum units; and Phase 3 packages the build for DepEd Learning Resource Portal (LRMDS) distribution.

> 💡 **Delivery Tip:** Walk through the 3 phases: USB offline installers -> teacher training -> DepEd LRMDS portal.

Q92. [Impact, Sustai] MEDIUM

How can Eco Quest adapt when local barangay sorting ordinances or national waste laws are amended?

Eco Quest is architected using ScriptableObject data containers (QuizData.asset, CityTrashDatabase.asset). Because waste categories, item definitions, and educational rationales are stored in modular data files rather than hardcoded in C# scripts, updating categories or adding new waste items requires only modifying asset data without rewriting core game code.

> 💡 **Delivery Tip:** Highlight that ScriptableObjects allow modifying trash and quiz rules without code rewrites.

Q93. [Impact, Sustai] HARD

What do you say to a panelist who claims: 'Eco Quest is just a video game that cannot solve the plastic crisis in the Philippines'?

We agree that software alone cannot solve infrastructure deficiencies like landfill capacity or industrial plastic production. However, Republic Act No. 9003 explicitly identifies public education and early habit formation as the mandatory first pillar of ecological waste management. Eco Quest equips the next generation with the cognitive instincts to segregate waste correctly, turning passive future citizens into active environmental stewards.

> 💡 **Delivery Tip:** Agree that software doesn't build landfills, but firmly argue that early education is the legally mandated first pillar.

Q94. [Impact, Sustai] MEDIUM

If your team had another six months and development funding, what top three features would you prioritize?

Our top three priorities would be: first, completing and linking World 3 (Beach Scene) to teach coastal and marine plastic conservation; second, developing an administrative teacher dashboard with local CSV score exports for grading; and third, implementing consumable waste generation in the Eco Shop so food purchases produce wrappers that must be sorted.

> 💡 **Delivery Tip:** List Beach Scene (World 3), teacher CSV dashboard, and consumable waste generation.

Q95. [Impact, Sustai] HARD

Why should STI College Sta. Mesa grant your team a passing grade for this capstone project today?

Because Eco Quest is not merely a proposal—it is a fully functional, rigorously tested 3D software application built to industry standards in Unity 6 and URP. It successfully translates complex environmental legislation (RA 9003) into engaging, kid-friendly gameplay, evaluated with 101 elementary pupils, backed by empirical data, and completely verified across code and execution.

> 💡 **Delivery Tip:** Deliver this closing defense pitch with confidence, pride, and clear professional conviction.

Q96. [Impact, Sustai] MEDIUM

How can Eco Quest be integrated into the Department of Education (DepEd) K-12 Science curriculum?

Eco Quest directly aligns with DepEd Grade 4 and Grade 5 Science standards regarding 'Changes in Materials' (biodegradable decomposition, recycling non-decaying materials, and the 5Rs). Teachers can assign World 1 as an interactive 15-minute laboratory activity to reinforce textbook units on solid waste classification.

> 💡 **Delivery Tip:** Cite the DepEd K-12 Science competency on 'Changes in Materials' and recycling non-decaying materials.

Q97. [Impact, Sustai] HARD

What was the single most difficult technical roadblock your team overcame during development?

Our greatest technical challenge was resolving physics desync and collider tunneling when pupils rapidly threw 3D trash props into bin triggers. We overcame this by engineering a custom multi-phase pickup pipeline in DragAndDrop.cs that combines a 5m SphereCast, kinematic socket lerping, NonAlloc overlap checks, and an automated settle failsafe that guarantees items never clip through the world.

> 💡 **Delivery Tip:** Describe the trash collider tunneling issue and how your DragAndDrop settle failsafe solved it.

Q98. [Impact, Sustai] EASY

What is the 30-second elevator pitch that summarizes the core value of Eco Quest?

Eco Quest is an interactive 3D educational desktop game that empowers Grade 4 to 6 pupils to master proper waste segregation through engaging experiential gameplay. Aligned with Republic Act No. 9003, it combines physical drag-and-drop sorting, pre- and post-quiz learning gain assessments, and an open City Hub to transform passive classroom environmental lessons into lifelong sustainable habits.

> 💡 **Delivery Tip:** Deliver this smoothly in under 30 seconds: 3D game, Grade 4-6, RA 9003, sorting, quizzes, lifelong habits.

Q99. [Impact, Sustai] HARD

What are the five strict operating rules your team will follow during this defense session?

Rule 1: Answer in one direct sentence first, then explain; Rule 2: Acknowledge project limitations before the panelist has to point them out; Rule 3: Never claim features or security that cannot be demonstrated in the build; Rule 4: Ensure every team member can explain the DFD, quest flow, and segregation code; Rule 5: Keep our high-definition gameplay video open as an immediate demo fallback.

> 💡 **Delivery Tip:** Memorize the five room rules and state them clearly to show exceptional team discipline.

Q100. [Impact, Sustai] HARD

In one clear sentence, what is the ultimate takeaway you want the panel to remember about Eco Quest?

Eco Quest proves that interactive technology can transform environmental education from theoretical classroom memorization into practical, engaging, and measurable waste segregation habits for the next generation.

> 💡 **Delivery Tip:** Speak this final sentence slowly, clearly, and look every panelist directly in the eye.

Defense Day Cheat Sheet — 30-Second Pitch & Facts to Know Cold

The 30-Second Elevator Pitch (Memorize This)

"Good morning, esteemed panel. **Eco Quest** is an interactive 3D educational desktop game designed for Grade 4 to 6 elementary pupils, built to transform solid waste segregation from passive classroom theory into practical, measurable habits. Aligned with Republic Act No. 9003, our system features physical drag-and-drop sorting, immediate feedback, pre- and post-quiz learning gain evaluations, and an open City Hub with citizen persuasion quests. Evaluated with 101 elementary pupils, Eco Quest proves that gamified simulation bridges the gap between environmental awareness and real-world sustainable action."

Key Technical Numbers to Know Cold

Metric	Exact Value to State	Where It Is Located in Code / Data
Target Audience	Grade 4 to 6 pupils (Ages 8–12)	101 evaluation respondents: 34.7% Gr4, 33.7% Gr5, 31.7% Gr6.
Waste Categories	3 in Forest (Bio, Recyc, Non-Bio) + 4 in City (+Hazard)	`QuestManager.cs`, `CityTrashDatabase.asset`
Penalty Amount	-10 Eco-Coins on 3rd consecutive mistake	`QuestManager.HandleMistake()` line 463; clamped at 0.
Ground Drop Penalty	Zero penalty (intentional child-friendly design)	`DragAndDrop.ReleaseObject()` line 2015.
Quiz Question Count	6 questions (Pre & Post identical pool, shuffled)	`QuizData.asset`; +5 gold reward per correct post-answer.
Eco Shop Catalog	9 consumable items (priced 15 to 55 coins)	`InventoryItemDatabase.asset` (Apples, Bread, Sandwiches, Drinks).
Daily Quest Cooldowns	5 mins (Ranger Robin) / 10 mins (Central Plaza Quiz)	`CityTrashDetectiveManager.cs` line 695; `LitterbugQuestManager.cs` line 1247.
Save Storage Path	Windows Registry (PlayerPrefs)	`HKCU\Software\DefaultCompany\EcoQuest`

Say vs Avoid Guide

✅ WHAT TO SAY:

- "The game provides *experiential simulation* that complements classroom lectures."
- "We implemented a *3-mistake forgiveness rule* to guide without frustrating young children."
- "Our evaluation measured *immediate cognitive knowledge acquisition*."
- "Settings and progress are stored safely on the *local Windows Registry*."
- "We prepared an addendum resolving the *12 missing bibliographic entries*."

❌ WHAT TO AVOID SAYING:

- *Do NOT claim:* "The game permanently changes real-world daily behavior."
- *Do NOT claim:* "The save system is completely secure and unhackable."
- *Do NOT claim:* "The daily quest uses an online midnight server reset."
- *Do NOT claim:* "Eco Quest replaces the school science teacher."
- *Do NOT say:* "That feature is coming next week" (say: "It is scoped for Version 2.0").

Five Operating Rules for the Defense Room — Tactical Defense Protocol

Rule 1: Answer in One Sentence First, Then Explain

Never start with "Because...", "Actually...", or long rambling stories. Give the direct conclusion in your first ten words: "*Yes, the penalty is 10 coins on the third mistake.*" Then provide the supporting technical rationale. Panelists appreciate clarity and directness.

Rule 2: Name Limitations Before the Panelist Does

If asked about behavior change, registry cheating, or daily quest clock bypass, admit it immediately: "*Yes, panelist, that is an inherent limitation of our offline local architecture.*" Acknowledging trade-offs proves you understand software engineering boundaries; defending a known flaw invites attacks.

Rule 3: Never Claim What You Cannot Demonstrate

Do not invent security algorithms, cloud databases, or AI systems that are not present in your C# scripts. If an ISO 25010 claim in the paper is an overclaim (e.g. Authenticity), clarify that assets are bundled in Unity binary formats rather than claiming cryptographic signatures.

Rule 4: Every Member Must Be Ready to Explain Core Systems

Panelists frequently target quieter team members with foundational questions: "*Explain DFD Level 0,*" "*How does the sorting quest validate tags?*", "*Where is save data stored?*" Every team member must practice their assigned technical deep-dive answers (Q10, Q60, Q61, Q62, Q63).

Rule 5: Keep a Recorded Gameplay Video as Instant Fallback

Live demos can freeze due to display resolution mismatches or projector HDMI handshake issues. Have a 1080p recorded video showing end-to-end gameplay (tutorial, sorting, quiz, city exploration) open in VLC Player minimized on the taskbar. If Unity stumbles, switch seamlessly to the video without panic.

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Good luck on November 14, 2026. Defend with confidence, evidence, and pride.